

Name of the Student	G.NAGA MANIKANTA
Reg. No	192425310
GitHub Link	

SIMATS ENGINEERING

CO5 - ASSESSMENT TOOL 1

Design Desk

COURSE NAME: Deep Learning
SLOT :

COURSE CODE: MLA04
Faculty : Dr.Arul Raj A.M

COURSE OUTCOME COVERED

CO5: Students will be able to analyze and explain advanced generative deep learning models including Autoencoders, Deep Belief Networks, Boltzmann Machines, Deep Boltzmann Machines, and Generative Adversarial Networks. They will be able to compare their architectures, explain their learning mechanisms, and apply suitable generative models to real-world problems. (BTL-3)

Course Outcome Weightage: 35%
Weightage: 30%

Duration: 90 minutes Assessment Tool

Mark : 25

Code of Conduct:

I G.NAGA MANIKANTA(192425310) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

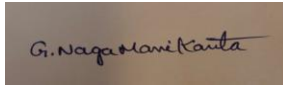
Signature of the student:

Name of the student: MANIKANTA

Criteria	Subject Knowledge and Conceptual Understanding (2)	Technical Accuracy and Application of Concepts (3)	Problem-Solving and Analytical Thinking (2)	Communication Skills and Clarity of Explanation (2)	Confidence, Presentation and Professional Behaviour (1)	Total (10)
Weightage	20 %	30%	20%	20 %	10 %	100%
Q1						

Q2						
Q3						
Q4						
Q5						
Total						

Faculty Signature :



Total Marks:

QUESTIONS: 25

Total Marks:25

1. Design an Autoencoder-Based Feature Learning System

A manufacturing company wants to detect defective products from high-dimensional sensor data. Design an Undercomplete Autoencoder with suitable encoder, latent representation, decoder, activation functions, and reconstruction loss. Explain how the design performs dimensionality reduction and feature learning.

ANSWER:

An **Undercomplete Autoencoder** reduces high-dimensional sensor data into a smaller latent representation.

Architecture:

Input → Encoder → Latent Space → Decoder → Reconstructed Output

Example:

100 → 64 → 20 → 64 → 100

- **Encoder:** Uses ReLU activation to extract important features.
- **Latent layer:** Contains fewer neurons, performing dimensionality reduction.
- **Decoder:** Reconstructs the original input.
- **Loss:** Mean Squared Error (MSE).

$$L = \frac{1}{n} \sum (x - \hat{x})^2$$

It learns important sensor patterns while removing redundant information, helping detect defective products.

2. Design a Robust Regularized Autoencoder

A healthcare organization has noisy patient-record features and needs a robust representation for downstream classification. Design a Regularized Autoencoder using an appropriate regularization strategy. Explain the encoder-decoder structure, regularization mechanism, training objective, and how the design reduces overfitting and improves useful feature extraction.

ANSWER:

A **Denoising Autoencoder** can be used for noisy patient data.

Architecture:

Noisy Input → Encoder → Latent Representation → Decoder → Clean Output

- Add noise to the input during training.
- Encoder uses ReLU to learn useful features.
- Decoder reconstructs the original clean data.
- Use **L2 regularization** to reduce overfitting.

$$L = L_{reconstruction} + \lambda || W ||^2$$

It learns robust features and becomes less sensitive to noise and irrelevant variations.

3. Design a Deep Generative Model

A recommendation or content-generation system requires learning complex data distributions. Design a Deep Belief Network or Deep Boltzmann Machine using suitable visible and hidden layers. Illustrate the architecture, explain the learning process, and justify the choice of the model for the proposed application.

ANSWER:

A **Deep Belief Network (DBN)** consists of multiple hidden layers.

Architecture:

Visible Layer $\rightarrow$ Hidden 1 $\rightarrow$ Hidden 2 $\rightarrow$ Hidden 3

For a recommendation system:

User-Item Data $\rightarrow$ 64 $\rightarrow$ 32 $\rightarrow$ 16

- First, layers are trained using **Restricted Boltzmann Machines (RBMs)**.
- Training is performed layer-by-layer.
- The network learns hierarchical features.
- Finally, the model is fine-tuned for recommendation or generation.

DBNs are suitable because they can learn complex patterns and hidden relationships in user preferences.

4. Design a Stochastic and Contractive Representation Model

A real-world image or signal-processing application requires representations that remain useful under small input variations. Design a model using Stochastic Encoders/Decoders or a Contractive Encoder. Specify the architecture, objective function, and robustness mechanism, and explain how the design improves representation stability.

ANSWER:

A **Contractive Autoencoder (CAE)** creates representations that are stable against small input changes.

Architecture:

Input Image $\rightarrow$ Encoder $\rightarrow$ Latent Space $\rightarrow$ Decoder $\rightarrow$ Output

- Encoder extracts compact features.
- Decoder reconstructs the image.
- A **contractive penalty** reduces sensitivity to small input variations.

$$L = L_{reconstruction} + \lambda \|J_f(x)\|_F^2$$

Thus, small changes such as noise or slight image shifts produce similar latent representations, improving robustness.

5. Design a GAN-Based Data Generation System

An organization has limited training images and wants to generate realistic synthetic samples. Design a Generative Adversarial Network with a Generator and Discriminator. Specify the input, network flow, adversarial training process, loss functions, and evaluation approach. Compare the proposed GAN design with an Autoencoder and justify why GAN is suitable for the application.

ANSWER:

A **GAN** consists of two networks:

- **Generator (G):** Creates fake images from random noise.
- **Discriminator (D):** Distinguishes real images from fake images.

Flow:

Random Noise → Generator → Fake Image → Discriminator

The discriminator also receives real images.

GAN training is adversarial:

$$\min_G \max_D E[\log D(x)] + E[\log(1 - D(G(z)))]$$

The Generator tries to fool the Discriminator, while the Discriminator tries to identify fake images.

GAN vs Autoencoder: GANs are better for generating realistic and sharp synthetic images, while autoencoders mainly focus on reconstruction and feature learning.

Design Desk Guidelines

- Identify the real-world problem, requirements, inputs, outputs and constraints.
- Draw a clear architecture or workflow diagram for the proposed model.
- Specify important layers, units, activation functions, latent representation and loss functions.
- Explain the training or learning mechanism and justify major design choices.
- Mention suitable evaluation measures and expected outcomes.
- Use neat labeling and present the design in a logical sequence.
- The solution should be original and written in the student's own words.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 Exemplary	Level 3 Proficient	Level 2 Developing	Level 1 Beginning
Understanding of Problem Context	20	Clearly understands the problem, requirements, constraints, and expected outcomes.	Good understanding with minor gaps in requirements.	Basic understanding; some requirements are unclear.	Poor understanding or misinterpretation of the problem.
Application of Concepts	30	Accurately applies appropriate generative deep learning concepts and architectures.	Applies relevant concepts correctly with minor errors.	Applies basic concepts with limited accuracy.	Fails to apply appropriate concepts.
Design Approach	20	Innovative, logical, well-structured, feasible, and technically justified design.	Logical and feasible design with minor gaps.	Basic design with some inconsistencies.	Unclear, illogical, or impractical design.
Accuracy of Solution	20	Completely correct architecture, process, objectives, and justification.	Mostly correct with minor technical errors.	Partially correct solution with noticeable gaps.	Incorrect or incomplete solution.
Clarity	10	Clear, well-organized diagram and explanation with proper technical labeling.	Understandable with minor clarity issues.	Limited clarity and explanation.	Poorly presented and difficult to understand.

